



Infobip USSD Gateway REST/JSON API v1.4

User manual

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1. INTRODUCTION

Infobip's USSD gateway has APIs designed to allow third party development of applications which can receive and send USSD messages. This manual document how a third party application developer can use one such API to integrate with Infobip's USSD gateway.

1.1. Intended Users

This API best suits third party application developers who:

1. Have the ability/resources/skills to host the entire application logic on their side
- all messages are merely forwarded, there is no logic or content manipulation on Infobip's USSD gateway.
2. Want to use Infobip's USSD service as quickly and simply as possible – as the API facilitates just message forwarding, it is the simplest to use and implement.

1.2. Naming

Some terms in this document have been shortened for clarity. Their full descriptions and abbreviated forms are listed in the following table:

Term	Abbreviated Term	Description
Infobip USSD Gateway	gateway	A software component which handles USSD communication between an application and the mobile network
Third-party USSD application	application	A software component which sends and receives USSD messages through the USSD gateway
Mobile number operator	MNO	Telecommunications service provider organization which provides data communication for its subscribed mobile users.

1.3. Sequence Diagram

Message flows between components are depicted with UML sequence diagrams. In case of gateway-application communication, requests and responses are actual HTTP requests and responses, where the gateway is the HTTP client and the application is the HTTP server.

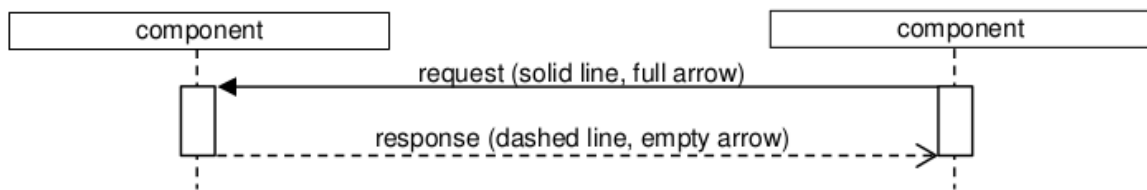


Figure 1 – an example of the UML sequence diagram

2. OVERVIEW

The REST/JSON API has three methods which application must implement to integrate with the gateway:

1. start – called to notify that a USSD session has been started and a first menu is needed to send to the mobile network,
2. response – called when a user replies to a USSD menu and the next menu is needed,
3. end – called when a session ends in case a last menu (also called a notification) was sent from the application, or an error occurs on the mobile network.

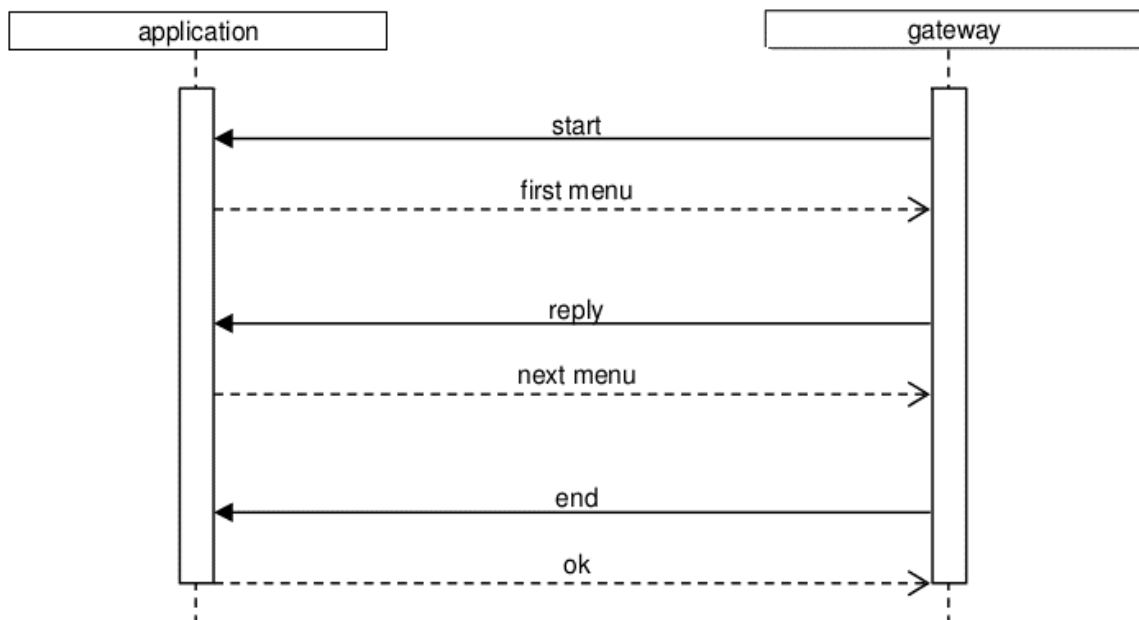


Figure 2 – REST/JSON API HTTP method requests and responses

2.1. Request Flows

There are two types of USSD sessions:

1. Notification – a text message is sent to the user's mobile phone without the possibility of user reply.
2. Dialog – a series of text messages are being exchanged between the application and the user's mobile phone, ending with a notification to the user.

2.1.1. Notification

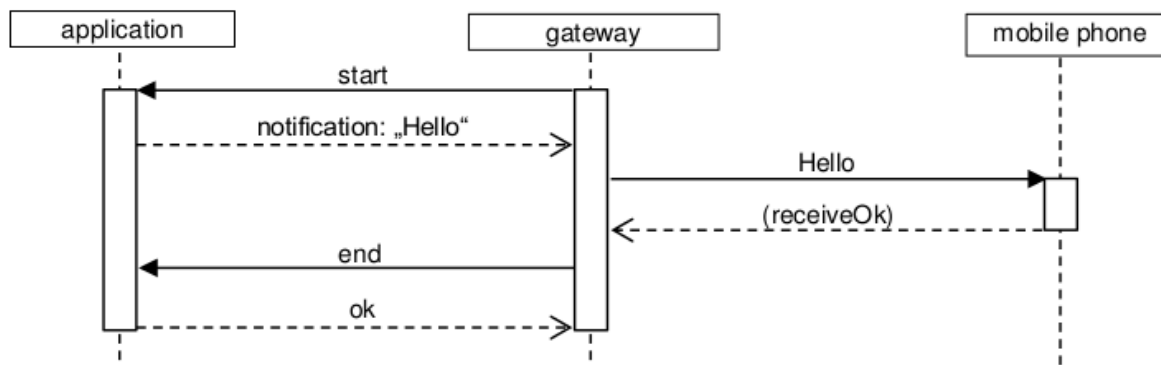


Figure 3 – an example of a USSD notification

2.1.2. Dialog

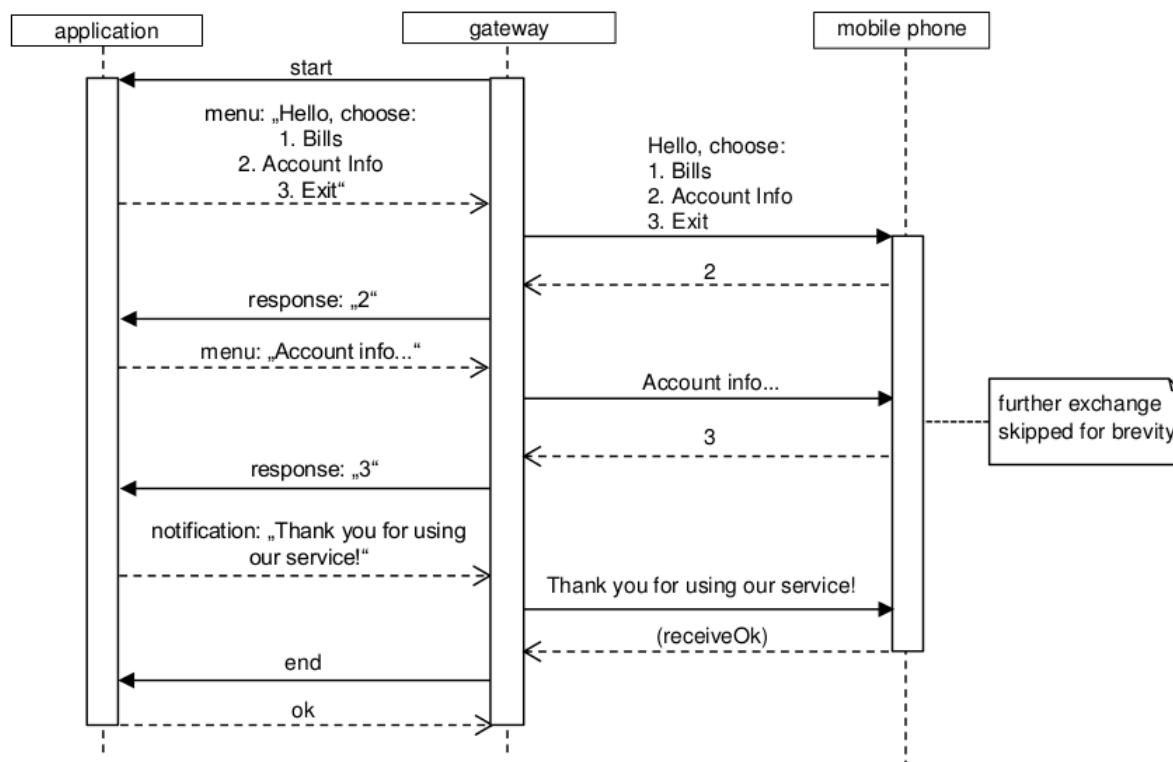


Figure 4 – an example of a USSD dialog

2.2. URL

The URL for calling the methods is generated from two parameters. Here are the URLs for each of the API methods:

```
{baseUrl}/session/{sessionId}/start  
  
{baseUrl}/session/{sessionId}/response  
  
{baseUrl}/session/{sessionId}/end
```

Figure 5 – HTTP method URL syntax

The `{baseUrl}` and `{sessionId}` placeholders are going to be filled with appropriate values by the gateway.

2.2.1. Base URL

This is the prefix of each URL. It is defined by the application developer and should be set under the „Base URL“ (`baseUrl`) property in the descriptor section of the application editor.

Include the entire HTTP URL on which your application will listen for requests, **but do not include a slash at the end.**

If the base URL is:

```
http://my.usssd.app:8081/restjsonApp
```

the URLs will look like:

```
http://my.usssd.app:8081/restjsonApp/session/{sessionId}/start
```

```
http://my.usssd.app:8081/restjsonApp/session/{sessionId}/response
```

```
http://my.usssd.app:8081/restjsonApp/session/{sessionId}/end
```

Following base URL is invalid, because it has a trailing slash:

```
http://my.usssd.app:8081/anotherApp/
```

2.2.2. Session ID

The session ID is a string of characters which is generated by the gateway at the start of each USSD session. Each session ID uniquely identifies the USSD session.

Session ID example:

```
13cc8b28afb86c69766531
```

URL examples:

```
http://my.ussd.app/restjsonApp/session/13cc8b28afb86c69766531/start
```

```
http://my.ussd.app/restjsonApp/session/13cc8b28afb86c69766531/response
```

```
http://my.ussd.app/restjsonApp/session/13cc8b28afb86c69766531/end
```

The application has to be able to parse the session ID from the HTTP request context path and use the ID to differentiate between sessions.

2.3. Authentication

It is possible for third party applications to secure their endpoints using Basic HTTP authentication. If this is the case, HTTP credentials should be configured within Infobip USSD platform. Then, USSD platform will add Authorization header to all HTTP requests towards third party applications.

3. METHOD REFERENCE

All HTTP requests contain the HTTP header: *Content-Type: application/json*

All HTTP requests contain the parameters in a JSON-encoded string in the request body.

All HTTP responses should contain the parameters in a JSON-encoded string in the response body.

3.1. Start

3.1.1. Request

Method	POST		
URL syntax	<code>{baseUrl}/session/{sessionId}/start</code>		
JSON parameters	Name	Description	Mandatory
	msisdn	user's phone number	x
	imsi	user's unique SIM card ID (supported only by several MNOs)	
	shortCode	the short code that was used in case of a mobile-initiated session	
	optional	auxiliary data that was sent when triggering the session	
	ussdNodeid	a unique ID of the gateway	
	text	auxiliary data that was sent when triggering the session	
	networkName	network to which end subscriber number belongs	
	countryName	country to which end subscriber numbers belongs	

3.1.2. Response

JSON parameters	Name	Description	Mandatory
	shouldClose	<i>true</i> if this is a notification (last menu), <i>false</i> otherwise	x
	ussdMenu	text of the first USSD menu, use \n for new line	x
	responseExitCode	see table in 4.1.	x
	responseMessage	specific description of the error, or an empty string	x

3.1.3. Example



Figure 6 – start method example

Note: to insert a new line in a menu, use the \n sequence.

3.2. Response

3.2.1. Request

Method	PUT		
URL syntax	<code>{baseUrl}/session/{sessionId}/response</code>		
JSON parameters	Name	Description	Mandatory
	msisdn	user's phone number	x
	imsi	user's unique SIM card ID (supported only by several MNOs)	
	shortCode	the short code that was used in case of a mobile-initiated session	
	optional	auxiliary data that was sent when triggering the session	
	ussdNodeIcd	a unique ID of the gateway	
	text	text of the user's reply to the menu	
	networkName	network to which end subscriber number belongs	
	countryName	country to which end subscriber numbers belongs	

3.2.2. Response

JSON parameters	Name	Description	Mandatory
	shouldClose	<i>true</i> if this is a notification (last menu), <i>false</i> otherwise	x
	ussdMenu	text of the first USSD menu, use \n for new line	x
	responseExitCode	see table in 4.1.	x
	responseMessage	specific description of the error, or an empty string	x

3.2.3. Example

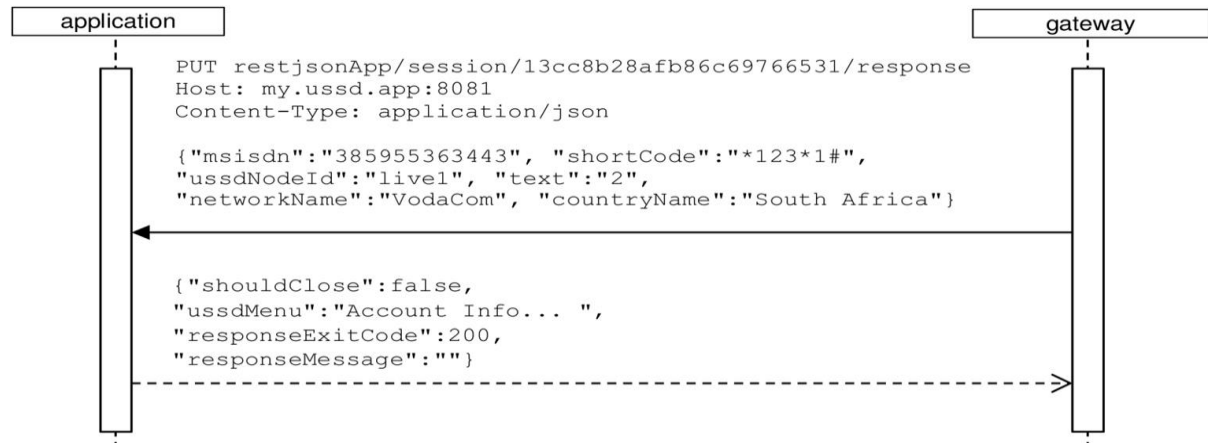


Figure 7 – response method example

3.3. End

3.3.1. Request

Method	PUT		
URL syntax	{baseUrl}/session/{sessionId}/end		
JSON parameters	Name	Description	Mandatory
	reason	a specific description of the error, or an empty string	x
	exitCode	see table in 4.2.	x
	networkName	network to which end subscriber number belongs	
	countryName	country to which end subscriber numbers belongs	

3.3.2. Response

JSON parameters	Name	Description	Mandatory
	responseExitCode	see table in 4.1.	x
	responseMessage	specific description of the error, or an empty string	x

3.3.3. Example



Figure 8 – end method example

4. APPENDIX

4.1. responseExitCode

Method calls are implemented through HTTP. Each HTTP response's JSON-encoded body should contain the parameter responseExitCode. Possible values are:

responseExitCode	Reason
200	Request processed without errors
400	Error processing JSON data
500	Internal server error

4.2. Session End

exitCode	Reason
200	Session ended normally
500	Session aborted by network
510	Session aborted by TPA
520	Session aborted by user
600	Session timeout

4.3. Previous versions of the document

4.3.1. Differences between 1.1 and 1.0

- Renamed *ussdGwId* to *ussdNodeId*

4.3.2. Differences between 1.2 and 1.1

- Delete method *Status* which was unused and unnecessary
- *End* method is triggering by PUT on
`{baseUrl}/session/{sessionId}/end`
- Delete *Language* request parameter which was unused and unnecessary
- Note that *IMSI* parameter does not makes any sense for some MNOs

4.3.2. Differences between 1.3 and 1.2

- Rebranding and redesign of the document

4.3.2. Differences between 1.4 and 1.3

- Added new section “2.3. Authentication”